Learning Notes: Two-Qubit Hamiltonian Simulation

Yutong Liu

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Abstract

These notes summarize the learning process behind the time evolution of a two-qubit spin-chain model. The project implements this model using two different methods: NumPy/SciPy exact statevector calculation and Qiskit circuit-based statevector simulation. Each script has inside numerical checks of normalization, initial expectation values, and symmetry relations. The Qiskit version additionally performs a cross-script numerical comparison of its circuit results with the NumPy matrix results.

This project aims to understand the running principles of the model, the role of each Hamiltonian term, the meaning and the time evolution of the observables, and the numerical checks in two implementations.

1 Project Content

This project studies the time evolution of a two-qubit spin-chain model using Hamiltonian simulation. The main content includes:

- Time evolution for a two-qubit quantum system based on a Hamiltonian
- Statevector representation of the four basis states
- Exact matrix-exponential simulation with NumPy/SciPy
- Statevector circuit simulation with Qiskit
- Numerical verification of the symmetry relations among the expectation value curves
- Numerical comparison between NumPy matrix results and Qiskit circuit results

The project system contains only two qubits, so each statevector has four components and the Hamiltonian is a 4×4 matrix, which makes the project easier for direct learning, numerical verification and reasoning step checking.

2 Two-Qubit State Representation

2.1 Computational Basis

The single-qubit basis states are $|0\rangle$ and $|1\rangle$. For two qubits, the computational basis states are

$$|00\rangle\ , \quad |01\rangle\ , \quad |10\rangle\ , \quad |11\rangle\ .$$

A general two-qubit state can be written as

$$|\psi\rangle = a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle\ , \quad \text{where } a\,, b\,, c\,, d \in \mathbb{C}$$

and the squared norms of these amplitudes $|a|^2$, $|b|^2$, $|c|^2$, and $|d|^2$ stand for the probabilities of measuring the corresponding basis states.

In this project, we suppose that all statevectors are normalized; hence we have:

$$|a|^2 + |b|^2 + |c|^2 + |d|^2 = 1\,.$$

In a two-qubit system, $|00\rangle$ and $|11\rangle$ have two qubits in the same direction; these two basis states are called **aligned states**, while $|01\rangle$ and $|10\rangle$ are called **anti-aligned states** because they have two qubits in the opposite directions.

2.2 Tensor-Product Operators

Two-qubit tensor product operators Let X and Z be the single-qubit Pauli matrices and let I be the identity operator.

For a two-qubit system, single-qubit operators are extended to the four-dimensional state space using tensor products. The same operators are written in three different ways below: as Qiskit labels, tensor products, and the notation used in these notes.

$$\texttt{ZI} = Z \otimes I =: Z_0\,, \qquad \texttt{IZ} = I \otimes Z =: Z_1\,, \qquad \texttt{ZZ} = Z \otimes Z =: Z_0 Z_1\,, \tag{2.1}$$

$$\texttt{XI} = X \otimes I =: X_0\,, \qquad \texttt{IX} = I \otimes X =: X_1\,. \tag{2.2}$$

In following sections, the Hamiltonian, state-evolution equations, and observables use this tensor-product notation.

Qubit-ordering convention Generally, Qiskit writes a two-qubit basis state as $|q_1 q_0\rangle$, where q_1 is the left bit, and q_0 is the right bit. The left character of a two-qubit Pauli label acts on q_1, while the right character acts on q_0, which are:

$$\texttt{XI/ZI} : X/Z \text{ acts on Qiskit } q_1\,, \qquad \texttt{IX/IZ} : X/Z \text{ acts on Qiskit } q_0\,.$$

This project uses a different labeling convention: the basis is written as $|q_0 q_1\rangle$, with q_0 on the left and q_1 on the right. Consequently, the same Qiskit Pauli labels are interpreted in this project as follows:

$$\texttt{XI/ZI} : X/Z \text{ acts on } q_0\,, \qquad \texttt{IX/IZ} : X/Z \text{ acts on } q_1\,.$$

Relative to Qiskit's convention, the labels in this project are mapped as

$$q_0 \,(\text{project}) \longleftrightarrow q_1 \,(\text{Qiskit})\,, \qquad q_1 \,(\text{project}) \longleftrightarrow q_0 \,(\text{Qiskit})\,.$$

For example, the basis state $|01\rangle$ is read by Qiskit as $q_1 = 0$ and $q_0 = 1$, while this project reads it as $q_0 = 0$ and $q_1 = 1$. Likewise, `ZI` applies Z to the left bit in both conventions, but in Qiskit's convention, Z acts on q_1; while in convention of this project, Z acts on q_0.

To summarize, in this project, Z_0 and X_0 act on the left qubit q_0, while Z_1 and X_1 act on the right qubit q_1.

3 Hamiltonian Model

In this project, the two-qubit spin-chain Hamiltonian is

$$H = J Z_0 Z_1 + h(X_0 + X_1)\,, \tag{3.1}$$

with parameters

$$J = 1.0\,, \quad h = 0.5\,,$$

where J is the interaction strength, determining how strongly the Hamiltonian distinguishes between aligned and anti-aligned cases, while h is the transverse-field strength, determining how strongly each qubit tends to flip.

The simulation in this project uses the convention $\hbar = 1$. In physical units, the Schrödinger equation would be written as

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H_{\text{phys}} |\psi(t)\rangle \ .$$

After dividing by $\hbar$, this becomes

$$i \frac{d}{dt} |\psi(t)\rangle = \frac{H_{\text{phys}}}{\hbar} |\psi(t)\rangle \ ,$$

and the Hamiltonian used in this simulation can be understood as

$$H = \frac{H_{\text{phys}}}{\hbar} \,.$$

For this reason, the parameters J and h have units of inverse time, while the time interval Δt, the spacing between adjacent time points, has the unit of time.

The project studies the four initial states

$$|00\rangle \ , \quad |01\rangle \ , \quad |10\rangle \ , \quad |11\rangle \ ,$$

on the time grid `np.linspace(0, 10, 200)`.

Denote $H_Z := JZ_0Z_1$ and $H_X := h(X_0 + X_1)$; then $H = H_Z + H_X$.

3.1 Hamiltonian Construction and State Evolution

The Hamiltonian in this project is time-independent. It does not have an explicit time dependence such as $H(t)$. Instead, a fixed Hamiltonian H generates a time-dependent state $|\psi(t)\rangle$.

Using the tensor-product operators from (2.2) and (2.1), the Hamiltonian can be written as

$$H = JZ_0Z_1 + h(X_0 + X_1) = J(Z \otimes Z) + h(X \otimes I + I \otimes X)\,,$$
$$H_Z = J(Z \otimes Z)\,, \qquad H_X = h(X \otimes I + I \otimes X)\,.$$

Because this is a two-qubit system, each tensor-product operator is a 4×4 matrix, so H is also a 4×4 Hermitian matrix.

With $\hbar = 1$, the state satisfies the time-dependent Schrödinger equation (TDSE)

$$i \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle \ . \tag{3.2}$$

For a time-independent Hamiltonian, the solution is

$$|\psi(t)\rangle = e^{-iHt} |\psi(0)\rangle \ . \tag{3.3}$$

Equivalently, the time-evolution operator is

$$U(t) = e^{-iHt}\,, \quad |\psi(t)\rangle = U(t) |\psi(0)\rangle \ . \tag{3.4}$$

In this project, this evolution is applied separately to each initial state, and the plotted curves are then calculated from these time-evolved states.

3.2 Role of the H_Z Term

For a single-qubit system, the Z operator leaves $|0\rangle$ unchanged and adds a minus sign to $|1\rangle$:

$$Z\,|0\rangle = +\,|0\rangle\ , \quad Z\,|1\rangle = -\,|1\rangle\ ,$$

so the Z_0Z_1 term defines the aligned and anti-aligned structure in the Z basis:

$$\begin{aligned} Z_0Z_1\,|00\rangle &= +\,|00\rangle\ , & Z_0Z_1\,|11\rangle &= +\,|11\rangle\ , \\ Z_0Z_1\,|01\rangle &= -\,|01\rangle\ , & Z_0Z_1\,|10\rangle &= -\,|10\rangle\ . \end{aligned}$$

It's clear that for aligned states $|00\rangle$ and $|11\rangle$, Z_0Z_1 assigns them the eigenvalue $+1$, while for anti-aligned states $|01\rangle$ and $|10\rangle$, Z_0Z_1 assigns them -1. As an observable, Z_0Z_1 distinguishes aligned and anti-aligned configurations in the Z basis. When included in the Hamiltonian as the term JZ_0Z_1, it represents a two-qubit interaction. Hence, Z_0Z_1 is a Z-direction correlation operator.

3.3 Role of the H_X Term

For a single-qubit system, the X gate (operator) swaps the $|0\rangle$ and $|1\rangle$ states:

$$X\,|0\rangle = |1\rangle\ , \quad X\,|1\rangle = |0\rangle\ .$$

So the $X_0 + X_1$ term drives amplitude flow among four basis states by flipping one qubit at a time. More precisely, for the initial state $|00\rangle$:

$$\begin{aligned} X_0\,|00\rangle = |10\rangle\ ,\ X_1\,|00\rangle = |01\rangle\ , \\ (X_0 + X_1)\,|00\rangle = |10\rangle + |01\rangle\ , \end{aligned}$$

and for other basis states:

$$\begin{aligned} (X_0 + X_1)\,|01\rangle &= |11\rangle + |00\rangle\ , \\ (X_0 + X_1)\,|10\rangle &= |00\rangle + |11\rangle\ , \\ (X_0 + X_1)\,|11\rangle &= |01\rangle + |10\rangle\ . \end{aligned}$$

This means that for an arbitrary statevector in a two-qubit system

$$|\psi\rangle = a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle\ ,$$

the H_X term connects the aligned pair $\{|00\rangle\ , |11\rangle\}$ with the anti-aligned pair $\{|10\rangle\ , |01\rangle\ \}$ by flipping one qubit at a time. It thereby transfers amplitude among the four basis states $|00\rangle\ , |01\rangle\ , |10\rangle\ , |11\rangle$ during time evolution.

4 Simulation Methods

4.1 NumPy Exact Matrix Simulation

The NumPy script realizes an exact calculation by directly constructing the Pauli matrices, tensor-product operators, Hamiltonian matrix, and time-evolution operator. For each time value in the chosen grid, it evaluates the same evolution rule described above:

$$U(t) = e^{-iHt}\,, \qquad |\psi(t)\rangle = U(t)\,|\psi(0)\rangle\ .$$

The results of NumPy script are encapsulated in a separate function which serves as a reference for Qiskit.

4.2 Qiskit Circuit Statevector Simulation

The Qiskit script reproduces the same expectation value curves of the NumPy version but in a circuit representation. In the circuit implementation, it represents the exact operator e^{-iHt} with a circuit evolution gate and explicitly selects matrix-exponential synthesis. It uses

- `SparsePauliOp` to define the Hamiltonian and observables
- `PauliEvolutionGate` to express time evolution as a circuit operation
- `MatrixExponential` as a synthesis option for the exact evolution gate
- `Statevector` to simulate the resulting circuit state

4.3 Numerical Validation

Both the NumPy and Qiskit versions perform several numerical checks with a tolerance of 10^{-10}:

- **Statevector normalization** For each initial basis state, the evolved statevector has a norm of 1 at every time point.
- **Initial expectation value** For $\langle Z_0 \rangle$, $\langle Z_1 \rangle$ and $\langle Z_0 Z_1 \rangle$, check whether these values at $t = 0$ match the expected ones for each initial basis state.
- **Symmetry** For each pair of Z expectation value curves related by symmetric relationships mentioned in Section 5.2, both scripts confirm these relations by checking whether the maximum absolute difference over all time points is under the numerical tolerance.

The Qiskit version also compares its three Z expectation values with the NumPy results for all four initial states at all time points. Each comparison requires a maximum absolute difference below 10^{-10}. After these checks pass, the script prints out the largest difference between the Qiskit and NumPy observable results across all time points.

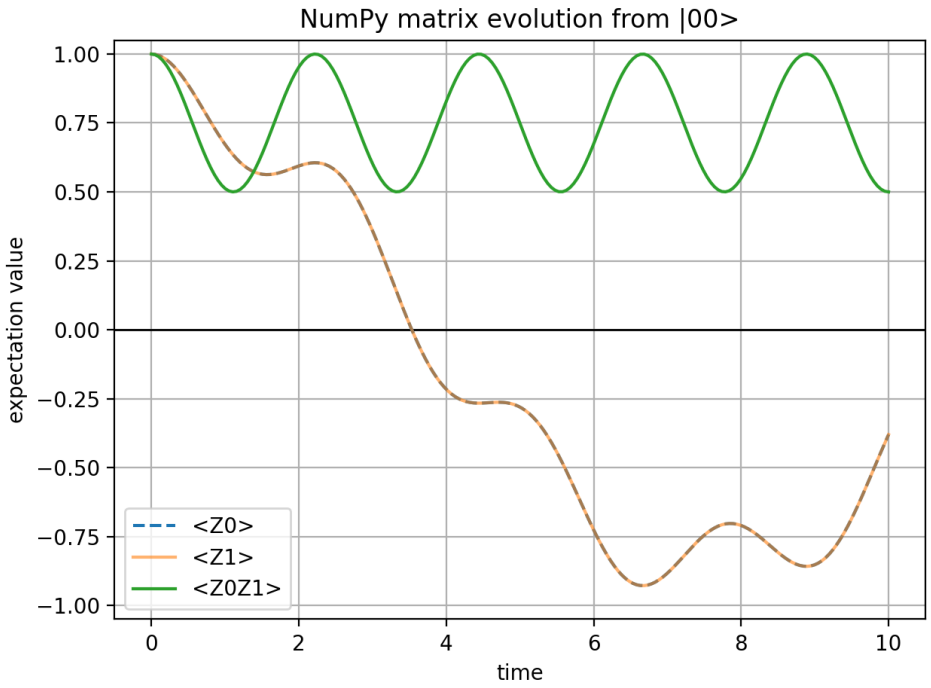
5 Observables and Interpretation

The project calculates three observables:

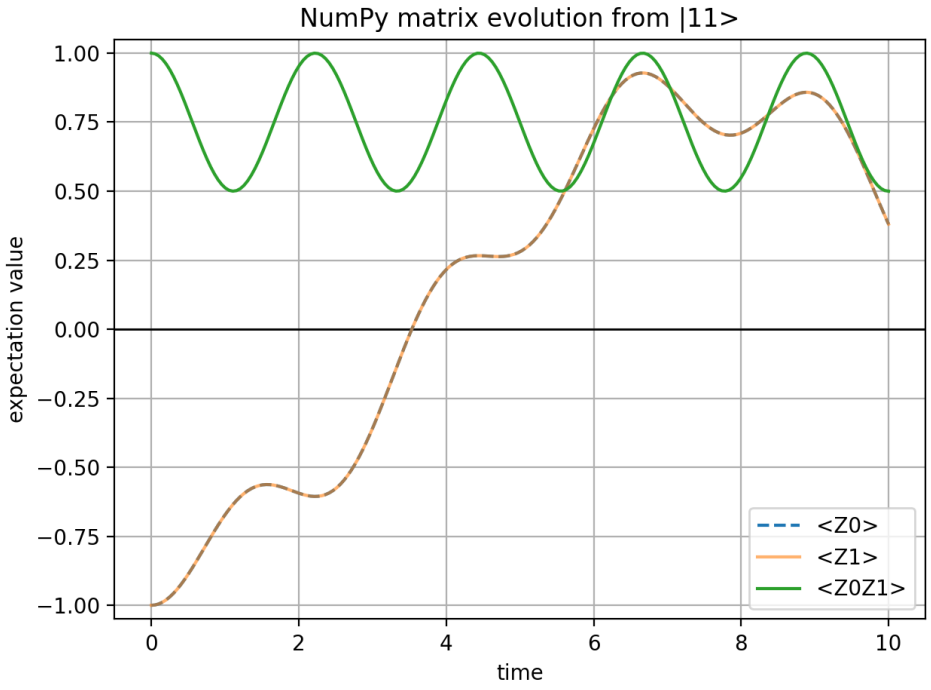
- $\langle Z_0 \rangle (t)$: the Z-direction expectation value of the left qubit q_0, measuring whether q_0 is biased toward $|0\rangle$ or $|1\rangle$
- $\langle Z_1 \rangle (t)$: the Z-direction expectation value of the right qubit q_1, measuring whether q_1 is biased toward $|0\rangle$ or $|1\rangle$
- $\langle Z_0 Z_1 \rangle (t)$: a two-qubit correlation observable, measuring whether the two qubits are biased toward being aligned or anti-aligned

The figures below are plots from the NumPy simulation, used to visualize the time evolution of the model and relationships among the observables.

Since these results from Qiskit agree with the NumPy reference within the numerical tolerance, only the NumPy plots are shown.

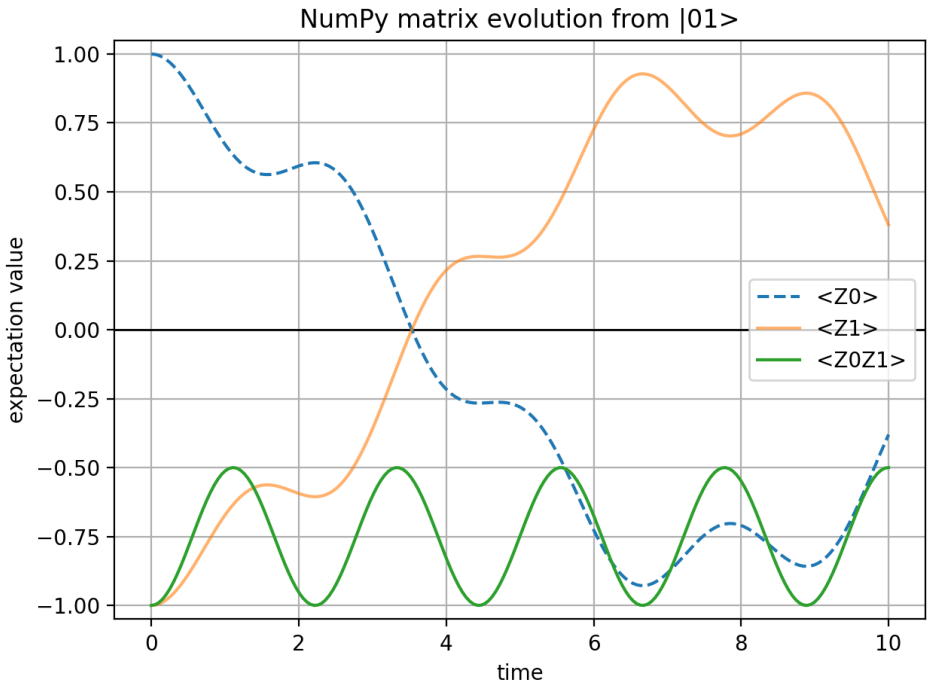


Initial state $|00\rangle$

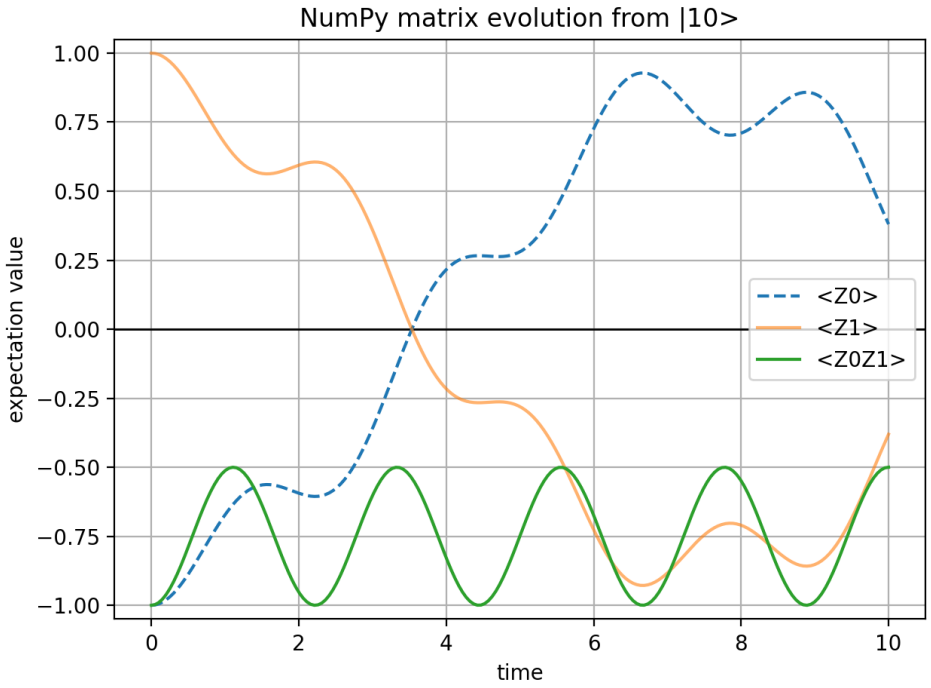


Initial state $|11\rangle$

Figure 1: Expectation value curves for aligned initial states $|00\rangle$ and $|11\rangle$.



Initial state $|01\rangle$



Initial state $|10\rangle$

Figure 2: Expectation value curves for anti-aligned initial states $|01\rangle$ and $|10\rangle$.

Figures 1 and 2 show how the same Hamiltonian produces different expectation value curves from different initial basis states.

Expectation value formulas For a general two-qubit time-evolved state

$$|\psi(t)\rangle = a(t)\,|00\rangle + b(t)\,|01\rangle + c(t)\,|10\rangle + d(t)\,|11\rangle \ ,$$

we calculate its three Z expectation values:

$$\begin{aligned}
\langle Z_0 \rangle (t) &= |a(t)|^2 + |b(t)|^2 - |c(t)|^2 - |d(t)|^2 \ , \\
\langle Z_1 \rangle (t) &= |a(t)|^2 - |b(t)|^2 + |c(t)|^2 - |d(t)|^2 \ , \\
\langle Z_0 Z_1 \rangle (t) &= |a(t)|^2 - |b(t)|^2 - |c(t)|^2 + |d(t)|^2 \ .
\end{aligned}$$

Define the **probability weight** of aligned states and anti-aligned states:

$$P_{\text{aligned}} := |a(t)|^2 + |d(t)|^2 \ , \quad P_{\text{anti-aligned}} := |b(t)|^2 + |c(t)|^2 \ ,$$

we can re-write $\langle Z_0 Z_1 \rangle$ as

$$\langle Z_0 Z_1 \rangle = P_{\text{aligned}} - P_{\text{anti-aligned}} \, . \tag{5.1}$$

When $\langle Z_0 Z_1 \rangle$ is positive, the two qubits are more likely to be aligned; and they are more likely to be anti-aligned when $\langle Z_0 Z_1 \rangle$ is negative.

5.1 Reasons for the Oscillatory Pattern of $\langle Z_0 Z_1 \rangle$

Reasons for the oscillation (5.1) replies that $\langle Z_0 Z_1 \rangle$ is the difference between the aligned and anti-aligned probability weights. So this value changes as the probability weight redistributed between the aligned states $\{|00\rangle \, , |11\rangle\}$ and the anti-aligned states $\{|01\rangle \, , |10\rangle\}$.

The transverse-field term

$$H_X = h(X_0 + X_1)$$

connects these two pairs by flipping one qubit at a time:

$$\{|00\rangle \, , \, |11\rangle\} \ \longleftrightarrow \ \{|01\rangle \, , \, |10\rangle\} \, .$$

Since the norm of statevectors remains 1 through the time evolution, we have

$$1 = |a(t)|^2 + |b(t)|^2 + |c(t)|^2 + |d(t)|^2 = P_{\text{aligned}} + P_{\text{anti-aligned}}$$

during this process, so P_{aligned} and $P_{\text{anti-aligned}}$ cannot increase or decrease at the same time. Moreover, $\langle Z_0 Z_1 \rangle$ equals the difference between these two probability weights, causing the expectation value to repeatedly decrease and increase following the oscillation of the pair of probability weights, producing an oscillatory curve.

Reasons for oscillating in sinusoidal form The Hamiltonian is a fixed 4×4 Hermitian matrix, and its energy eigenstates satisfy the time-independent Schrödinger equation (TISE):

$$H\,|E_k\rangle = E_k\,|E_k\rangle \ . \tag{5.2}$$

Expand the initial state as

$$|\psi(0)\rangle = \sum_k c_k\,|E_k\rangle \ ,$$

then its time evolution is

$$|\psi(t)\rangle = e^{-iHt}\,|\psi(0)\rangle = \sum_k c_k e^{-iE_k t}\,|E_k\rangle\ .$$

Substituting this expansion into the expectation value gives

$$\langle Z_0 Z_1\rangle(t) = \sum_{m,n} c_m^* c_n e^{i(E_m - E_n)t}\,\langle E_m|\,Z_0 Z_1\,|E_n\rangle\ .$$

The factor $e^{iE_m t}$ comes from $\langle\psi(t)|$, while $e^{-iE_n t}$ comes from $|\psi(t)\rangle$, leading to time-dependent terms with phase products of the form:

$$e^{iE_m t}e^{-iE_n t} = e^{-i(E_n - E_m)t}\,.$$

Using Euler's formula,

$$e^{-i\omega t} = \cos(\omega t) - i\sin(\omega t)\,,$$

these phase products become sine- and cosine-type terms whose frequencies are determined by the energy differences

$$\omega_{mn} = E_n - E_m\,.$$

For these reasons, $\langle Z_0 Z_1\rangle(t)$ is a finite combination of sine- and cosine-type terms. For the present initial states and observable, this combination produces the regular sinusoidal-type pattern seen in the numerical plots.

5.2 Symmetry and Position of the Curves

This subsection explains the following phenomena of the four expectation value plots above:

1. For aligned initial states, the curves of $\langle Z_0\rangle$ and $\langle Z_1\rangle$ coincide within each plot. When the $|00\rangle$ and $|11\rangle$ plots are compared, the corresponding single-qubit curves are reflections of each other across the horizontal axis.

2. For anti-aligned initial states, the $\langle Z_0\rangle$ and $\langle Z_1\rangle$ curves are reflections of each other across the horizontal axis within each plot. When the $|01\rangle$ and $|10\rangle$ plots are compared, the curves of the same single-qubit observable are also reflections of each other across the horizontal axis.

3. In the plotted parameter setting, the $\langle Z_0 Z_1\rangle$ curves remain above the zero-expectation axis for aligned initial states and below it for anti-aligned initial states. The aligned and anti-aligned curves are related by horizontal-axis reflection.

Here "horizontal axis" refers to the zero-expectation line $\langle O\rangle = 0$, and "horizontal axis reflection" means sign reversal across the horizontal axis, rather than a reversal of the time direction.

5.2.1 Aligned initial states: $|00\rangle$ and $|11\rangle$

Coincidence of the $\langle Z_0\rangle$ and $\langle Z_1\rangle$ curves For an evolved two-qubit state $|\psi(t)\rangle$ whose initial state is aligned ($|00\rangle$ or $|11\rangle$), we calculate its $\langle Z_0\rangle$ and $\langle Z_1\rangle$. Take the case where the initial state is $|00\rangle$, denote

$$|\psi_{00}(t)\rangle \quad \text{with} \quad |\psi_{00}(0)\rangle = |00\rangle \quad \text{when} \quad t = 0\,,$$
$$\langle Z_0\rangle_{00}(t) := \langle\psi_{00}(t)|\,Z_0\,|\psi_{00}(t)\rangle\ , \quad \langle Z_0\rangle_{00}(0) = \langle\psi_{00}(0)|\,Z_0\,|\psi_{00}(0)\rangle = \langle 00|\,Z_0\,|00\rangle\ ,$$
$$\langle Z_1\rangle_{00}(0) := \langle\psi_{00}(0)|\,Z_1\,|\psi_{00}(0)\rangle = \langle 00|\,Z_1\,|00\rangle\ .$$

Then we obtain:

$$Z_0 \left|00\right\rangle = + \left|00\right\rangle \ , \ \left\langle Z_0 \right\rangle_{00} (0) = + \left\langle 00|00\right\rangle = +1 \, ,$$
$$Z_1 \left|00\right\rangle = + \left|00\right\rangle \ , \ \left\langle Z_1 \right\rangle_{00} (0) = + \left\langle 00|00\right\rangle = +1 \, .$$

Hence, we observe that the curves of $\langle Z_0 \rangle$ and $\langle Z_1 \rangle$ start at the same point $(0\,,\,+1)$.

To understand why the two curves continue to coincide during the whole time evolution, we then look at the time-dependent amplitudes directly.

Substituting $|\psi(t)\rangle$ into the TDSE gives the amplitude equations:

$$i\dot{a} = Ja + hb + hc \, ,$$
$$i\dot{b} = ha - Jb + hd \, ,$$
$$i\dot{c} = ha - Jc + hd \, ,$$
$$i\dot{d} = hb + hc + Jd \, .$$

We put two differential equations of $b(t)$ and $c(t)$ together for comparison:

$$i\dot{b} = ha - Jb + hd = -Jb + h(a + d) \, ,$$
$$i\dot{c} = ha - Jc + hd = -Jc + h(a + d) \, ,$$

which have the same differential equation structure:

$$i\dot{x} = -Jx + h\left(a(t) + d(t)\right) \, .$$

In the case that $|00\rangle$ is the initial state, these two amplitudes start from the same value:

$$b(0) = c(0) = 0 \, .$$

Moreover, $b(t)$ and $c(t)$ satisfy the same 1st-order ODE with the same initial value, so they remain equal throughout the time evolution, that is:

$$b(t) = c(t) \quad \forall \, t \geq 0 \, .$$

As a result,

$$\left|b(t)\right|^2 = \left|c(t)\right|^2 \, .$$

It follows that

$$\begin{aligned} \left\langle Z_0 \right\rangle (t) - \left\langle Z_1 \right\rangle (t) &= |a(t)|^2 + |b(t)|^2 - |c(t)|^2 - |d(t)|^2 - \left(|a(t)|^2 - |b(t)|^2 + |c(t)|^2 - |d(t)|^2 \right) \\ &= 2 \left(|b(t)|^2 - |c(t)|^2 \right) = 0 \, , \end{aligned}$$

so finally we get

$$\left\langle Z_0 \right\rangle (t) = \left\langle Z_1 \right\rangle (t) \quad \forall \, t \geq 0 \, ,$$

which clarifies why the two curves both start at $(0\,,\,+1)$ and coincide throughout the whole time evolution.

The same argument applies to the initial state $|11\rangle$. In this case,

$$a(0) = b(0) = c(0) = 0 \, , \quad d(0) = 1 \, ,$$

so we still have $b(0) = c(0)$. Since $b(t)$ and $c(t)$ obey the same evolution rule and start at the same initial value, they remain equal throughout. As a consequence:

$$\left\langle Z_0 \right\rangle_{11} (t) = \left\langle Z_1 \right\rangle_{11} (t) \quad \forall \, t \geq 0 \, . \tag{5.3}$$

The two curves start from -1 rather than $+1$.

Horizontal-axis reflection between $|00\rangle$ and $|11\rangle$ The operator X_0X_1 maps $|00\rangle$ to $|11\rangle$ by flipping both two qubits at one time and leaves the Hamiltonian invariant:

$$(X_0X_1)H(X_0X_1) = H\,.$$

Under the same operation, the single-qubit Z observables change sign:

$$(X_0X_1)Z_0(X_0X_1) = -Z_0\,, \quad (X_0X_1)Z_1(X_0X_1) = -Z_1\,.$$

Thus the coincident single-qubit curves for $|11\rangle$ are the sign-reversed versions of the corresponding curves for $|00\rangle$:

$$\langle Z_0\rangle_{11}(t) = -\langle Z_0\rangle_{00}(t)\,, \quad \langle Z_1\rangle_{11}(t) = -\langle Z_1\rangle_{00}(t)\,. \tag{5.4}$$

This is the cross-plot horizontal-axis reflection visible between the $|00\rangle$ and $|11\rangle$ figures.

5.2.2 Anti-aligned initial states: $|01\rangle$ and $|10\rangle$

Horizontal-axis reflection of $\langle Z_0\rangle$ and $\langle Z_1\rangle$ within each plot For the anti-aligned case, take the initial state $|01\rangle$ as the example. We use the same notation as in the $|00\rangle$ case:

$$\begin{aligned} Z_0|01\rangle &= +|01\rangle\,, \quad \langle Z_0\rangle_{01}(0) = +\langle 01|01\rangle = +1\,, \\ Z_1|01\rangle &= -|01\rangle\,, \quad \langle Z_1\rangle_{01}(0) = -\langle 01|01\rangle = -1\,. \end{aligned}$$

Thus $\langle Z_0\rangle_{01}$ and $\langle Z_1\rangle_{01}$ start at opposite points $(0\,,+1)$ and $(0\,,-1)$.

Then we look at time-dependent amplitudes. For the initial state $|01\rangle$, the two amplitudes relevant to the within-plot sign symmetry are $a(t)$ and $d(t)$, and they start from the same value:

$$a(0) = d(0) = 0\,.$$

Their equations of motion are

$$\begin{aligned} i\dot{a} &= Ja + h(b+c)\,, \\ i\dot{d} &= Jd + h(b+c)\,, \end{aligned}$$

which have the same differential equation structure:

$$i\dot{x} = Jx + h\left(b(t) + c(t)\right)\,.$$

$a(t)$ and $d(t)$ satisfy the same 1st-order ODE with the same initial value, so they remain equal throughout the time evolution, and their squared magnitudes remain equal as well:

$$a(t) = d(t) \quad \Rightarrow \quad |a(t)|^2 = |d(t)|^2 \quad \forall\, t \geq 0\,.$$

It follows that

$$\begin{aligned} \langle Z_0\rangle_{01}(t) + \langle Z_1\rangle_{01}(t) &= |a(t)|^2 + |b(t)|^2 - |c(t)|^2 - |d(t)|^2 + \left(|a(t)|^2 - |b(t)|^2 + |c(t)|^2 - |d(t)|^2\right) \\ &= 2\left(|a(t)|^2 - |d(t)|^2\right) = 0\,. \end{aligned}$$

Therefore,

$$\langle Z_0\rangle_{01}(t) = -\langle Z_1\rangle_{01}(t) \quad \forall\, t \geq 0\,. \tag{5.5}$$

This relation explains the reason that two single-qubit curves in the $|01\rangle$ plot are reflected across the horizontal zero-expectation axis throughout the whole time evolution.

For $|10\rangle$, the initially nonzero amplitude is $c(0) = 1$ instead of $b(0) = 1$, but as the key condition $a(0) = d(0) = 0$ still holds, the same reasoning also works for $|10\rangle$ case.

Cross-plot reflection between $|01\rangle$ and $|10\rangle$ The Hamiltonian is invariant under exchanging qubit 0 and qubit 1. This exchange maps $|01\rangle$ to $|10\rangle$ and interchanges Z_0 with Z_1. Hence,

$$\langle Z_0\rangle_{01}(t) = \langle Z_1\rangle_{10}(t)\,, \quad \langle Z_1\rangle_{01}(t) = \langle Z_0\rangle_{10}(t)\,.$$

From the previous paragraph, we have

$$\langle Z_0\rangle_{01}(t) = -\langle Z_1\rangle_{01}(t)\,, \quad \langle Z_0\rangle_{10}(t) = -\langle Z_1\rangle_{10}(t)\,.$$

Combining these relations gives:

$$\langle Z_0\rangle_{10}(t) = -\langle Z_0\rangle_{01}(t)\,, \quad \langle Z_1\rangle_{10}(t) = -\langle Z_1\rangle_{01}(t)\,. \tag{5.6}$$

5.2.3 Symmetry Relations of the $\langle Z_0Z_1\rangle$ Curves

The initial values separate the aligned and anti-aligned states:

$$\langle Z_0Z_1\rangle_{00}(0) = \langle Z_0Z_1\rangle_{11}(0) = +1\,,$$
$$\langle Z_0Z_1\rangle_{01}(0) = \langle Z_0Z_1\rangle_{10}(0) = -1\,.$$

Thus the $\langle Z_0Z_1\rangle$ curves of aligned initial states begin in the upper region of the plots, whereas the curves of anti-aligned initial states begin in the lower region.

Coincidence of correlation curves within the aligned and anti-aligned pairs The operator X_0X_1 maps the basis initial states as follows:

$$(X_0X_1)\,|00\rangle = |11\rangle\ , \quad (X_0X_1)\,|01\rangle = |10\rangle\ ,$$

and it leaves both the Hamiltonian and the Z_0Z_1 operator unchanged:

$$(X_0X_1)H(X_0X_1) = H\,, \quad (X_0X_1)Z_0Z_1(X_0X_1) = Z_0Z_1\,.$$

Since X_0X_1 leaves H invariant, it also commutes with $U(t) = e^{-iHt}$. As a result, this operator keeps the time evolution and $\langle Z_0Z_1\rangle$ unchanged while mapping $|00\rangle$ to $|11\rangle$ and $|01\rangle$ to $|10\rangle$. The two aligned initial states consequently have the same curve of $\langle Z_0Z_1\rangle$, as do the anti-aligned initial states:

$$\langle Z_0Z_1\rangle_{00}(t) = \langle Z_0Z_1\rangle_{11}(t)\,, \tag{5.7a}$$
$$\langle Z_0Z_1\rangle_{01}(t) = \langle Z_0Z_1\rangle_{10}(t)\,. \tag{5.7b}$$

Horizontal-axis reflection between aligned and anti-aligned curves For each basis initial state $|s\rangle$, the time-evolved state is

$$|\psi_s(t)\rangle = U(t)\,|s\rangle\ ,\ s \in \{00, 01, 10, 11\}\,,$$

and its correlation curve $\langle Z_0Z_1\rangle$ is defined as

$$f_s(t) := \langle Z_0Z_1\rangle_s(t) = \langle\psi_s(t)|\,Z_0Z_1\,|\psi_s(t)\rangle = \langle s|\,U^\dagger(t)Z_0Z_1U(t)\,|s\rangle\ , \quad s \in \{00, 01, 10, 11\}\,.$$

Each $f_s(t)$ is a diagonal matrix element of $U^\dagger(t)Z_0Z_1U(t)$ in the basis. Therefore, summing over the four basis states gives the trace:

$$\sum_s f_s(t) = f_{00}(t) + f_{01}(t) + f_{10}(t) + f_{11}(t) = \mathrm{Tr}\left(U^\dagger(t)Z_0Z_1U(t)\right)\,.$$

Applying the cyclic property of trace and the unitarity relation $U(t)U^\dagger(t) = I$, we have

$$\mathrm{Tr}\left(U^\dagger(t)Z_0Z_1U(t)\right) = \mathrm{Tr}\left(Z_0Z_1U(t)U^\dagger(t)\right) = \mathrm{Tr}(Z_0Z_1)\,.$$

In the basis $\{|00\rangle\,, |01\rangle\,, |10\rangle\,, |11\rangle\}$, the diagonal entries of Z_0Z_1 are $+1, -1, -1, +1$. Hence,

$$\mathrm{Tr}(Z_0Z_1) = (+1) + (-1) + (-1) + (+1) = 0\,,$$
$$\sum_s f_s(t) = f_{00}(t) + f_{01}(t) + f_{10}(t) + f_{11}(t) = 0\,.$$

Since $f_s(t) := \langle Z_0Z_1\rangle_s(t)$, combining (5.7a) and (5.7b) gives

$$f_{00}(t) = f_{11}(t)\,, \quad f_{01}(t) = f_{10}(t)\,.$$

Substituting these relations above into the sum, we have

$$0 = f_{00}(t) + f_{01}(t) + f_{10}(t) + f_{11}(t) = 2f_{00}(t) + 2f_{01}(t) = 2f_{11}(t) + 2f_{10}(t)\,.$$

Therefore,

$$f_{01}(t) = -f_{00}(t)\,, \quad f_{10}(t) = -f_{11}(t)\,.$$

Finally, returning to the expectation value notation, we obtain

$$\langle Z_0Z_1\rangle_{01}(t) = -\langle Z_0Z_1\rangle_{00}(t)\,, \tag{5.8a}$$
$$\langle Z_0Z_1\rangle_{10}(t) = -\langle Z_0Z_1\rangle_{11}(t)\,. \tag{5.8b}$$

These relations describe the horizontal-axis reflection of $\langle Z_0Z_1\rangle$ curve between the aligned and anti-aligned initial states. In the plotted parameter setting, the numerical results also show that the aligned curves remain above the horizontal axis and the anti-aligned curves remain below it.

6 Scope and Possible Extensions

The completed scope is:

$$\begin{array}{ccc} \text{NumPy exact matrix simulation} & \longrightarrow & \text{NumPy plots} \\ \Big\uparrow \text{numerical comparison} & \Longrightarrow & \text{same plots} \Big\updownarrow \\ \text{Qiskit circuit statevector simulation} & \longrightarrow & \text{Qiskit plots} \end{array}$$

Possible local extensions include:

- Trotter error analysis for comparing exact evolution with product-formula approximations
- Parameter sweeps over different J, h, and total evolution time
- Extension from two qubits to small spin chains
- Additional observables, such as entanglement entropy or state fidelity

7 Summary

This project studies the time evolution of a two-qubit spin-chain model. Mainly focusing on the three expectation values and relations among their curves, this project observes and explains their behaviors by expressing these expectation values in terms of basis-state probabilities.

Meanwhile, the NumPy and Qiskit implementations compute the time evolution of the variables with numerical checks and curve plots, providing an intuitive view of operating principles behind the model and relationships among the variables.